

- a) **3:06.36** b) 3:03.66 c) 3:36.06 d) 3:30.66
45. This is the amount of a property in which a willing buyer will pay to a willing seller for the property when neither one is under the compulsion to buy or sell. What do you call this value?
a) Fair value b) Goodwill value c) Book value **d) Market value**
46. Determine the diameter of a circle, $x^2 + y^2 - 6x + 4y - 12 = 0$.
a) 9 units b) 11 units c) 12 units **d) 10 units**
47. What is the minimum point of $y = x + \frac{1}{x}$?
a) (1, 2) b) (1.5, 2) c) (1, 1.5) d) (2, 1)
48. The volume of the cube is increasing at a rate of 5 cu. m per minute. Determine the rate at which the surface area is increasing, in m^2/min , when its side is 10 m.
a) $\frac{1}{2}$ b) 1.75 c) 1.5 **d) 2**
49. What is the simple interest rate if an investment of P37,500.00 accumulates to P45,937.5 in 18 months?
a) 0.15 b) 0.2 c) 0.21 d) 0.3
50. What do you think is the negotiation of wage rates, conditions of employment, etc. by representatives of the labor force and management?
a) Union trade b) Union rally **c) Collective bargaining** d) Cooperative
51. What is the value of $\lim_{x \rightarrow \infty} \frac{x^2 - 1}{x^3 - 1}$?
a) 0 b) 0.25 c) 1.25 d) indeterminate
52. What is the present value of P5,000.00 due in 8 years if money is worth 12% compounded semi-annually?
a) P1,896.23 b) P1,869.23 c) P1,698.23 **d) P1,968.23**
53. This is a type of bond whose guaranty is in lieu on railroad equipment. What is this type of bond?
a) Equipment b) Debenture bond c) Registered bond **d) Infrastructure bond**
54. How do you call a type of bond where the corporation's owner name is recorded and the interest is paid periodically to the owners with their asking for it?
a) Registered bond b) Preferred bond c) Incorporator's bond d) Bail bond
55. A man left Sta. Rosa City to drive to Lopez, Quezon at 6:15 pm and arrived at 11:45 pm. If he averaged 50 kph and stopped 1 hour for dinner, how far is Lopez, quezon from Sta. Rosa City?
a) 225 km b) 522 km **c) 252 km** d) 215 km
56. If x varies directly as y and inversely as z, and x = 14 when y = 7 and z = 2, find the value of x when z = 4 and y = 16.
a) 12 b) 18 c) 14 **d) 16**
57. How many terms of the sequence – 9, - 6, - 3, ... must be taken so that the sum is 66?
a) 11 b) 6 c) 4 d) 9
58. There are 6 geometric means between 4 and 8748. Find the sum of all terms.
a) 12 310 b) 12 130 c) 13 210 **d) 13 120**
59. How many permutation can be made out of the letters of the word ENGINEERING?
a) 277 200 b) 272 700 c) 200 277 d) 227 700
60. A bag contains 3 white and 5 red balls. If two balls are drawn at random, find the probability that both are white.
a) 0.375 **b) 0.10714** c) 0.46667 d) 0.28571
61. Determine the absolute value of the complex number $3 + 4i$.
a) 4 **b) 5** c) 8 d) 6
62. Simplify $i^{1997} + i^{1999}$, where i is an imaginary number.
a) 0 b) i c) $1 + i$ d) $1 - i$
63. The sum of the two interior angles of the triangle is equal to the third angle and the difference of the two angles is equal to $\frac{2}{3}$ of the 3rd angle. Find the 3rd angle.
a) 60° b) 30° **c) 90°** d) 40°
64. If vector A is 10 units and vector B, which makes 60° with vector A, is 20 units. What is the difference of vectors A and B?
a) 15.34 units b) 13.45 units c) 18.76 units **d) 17.32 units**
65. How do you call the integral of any quotient whose numerator is the differential of the denominator?
a) Co-logarithm **b) Logarithm** c) Product d) Derivative
66. What is a regular polygon that has 27 diagonals?
a) Nonagon b) hexagon c) Pentagon d) Heptagon
67. How do you call the formula used to compute the value of n factorial, which is in symbolic form (n!), where n is large number?
a) Matheson formula b) Diophantine formula c) Richardson-Duchman formula **d) Stirling's Approximation**
68. What is the reason why an ivory soap floats in water?

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- a) All matter has mass
c) The density of ivory soap is unity
b) The specific gravity of ivory soap is greater than that of water
d) The specific gravity of ivory soap is less than that of water
69. When two planes intersect with each other, the amount of divergence between the two planes is expressed by measuring the:
a) Reflex angle **b) Dihedral angle** c) Polyhedral angle d) Plane angle
70. What do you think is the output or sales at which income is insufficient to equal operating cost?
a) Break even point b) Depreciation c) Investment d) Cash flow
71. What is an estimate of assets' net market value at the end of its estimated life?
a) Book value b) Depreciation c) **Salvage value** d) Cash flow
72. What do you think is the lessening of the value of an asset due to a decrease in the quantity available as a coal, oil and timber in forests?
a) Depletion b) Amortization c) Depreciation d) Investment
73. What can you say about the present worth of all depreciation over the economic life of the item?
a) Maintenance b) Capital recovery c) **Depreciation recovery** d) Annuity
74. What do you think is the provision in the contract that indicates the possible adjustment of material cost and labor cost?
a) Secondary clause b) Specification c) **Escalatory clause** d) General provision
75. This is the process of determining the value of certain property for specific reasons. Guess, what is this?
a) Amortization b) Investment c) **Appraisal** d) Depreciation
76. How do you call those products or services that are directly used by people to satisfy their wants?
a) **Consumer goods and services** b) Producer goods and services
c) Necessity products and services d) Luxury products and services
77. These are used to produce consumer goods and services. Guess, what are these?
a) Consumer goods and services b) **Producer goods and services**
c) Necessity products and services d) Luxury products and services
78. What do you think are those products or services that are required to support human life and activities that will be purchased in somewhat the same quantity even though the price varies considerably?
a) Consumer goods and services b) Producer goods and services
c) **Necessity products and services** d) Luxury products and services
79. How do you call a cylinder with elliptical cross section?
a. Ellipsoid b. **Cylindroid** c. Hyperboloid d. Paraboloid
80. How do you call a market whereby there is only one buyer of an item for which there are no goods substitutes?
a) Monopoly **b) Monopsony** c) Oligopoly d) Oligopsony
81. When a line $y = mx + b$ slopes downwards from left to right, the slope m is:
a) Less than zero b) Greater than zero c) Equal to zero d) Equal to 1
82. "Whenever a net force acts on a body, it produces acceleration in the direction of the resultant force, an acceleration which is directly proportional to the resultant force and inversely proportional to the resultant force and inversely proportional to the mass of the body". How do you call this theory?
a) Newton's First law of motion b) **Newton's Second Law of motion**
c) Faraday's law of forces d) Hook's law of equilibrium
83. It is defined that the momentum of a moving object is the product of its mass, m , and velocity, V . In Newton's Law of Motion, what is the rate of change of momentum with respect to time?
a) Power b) Momentum c) energy **d) Force**
84. The loss of weight of a body submerged in a fluid is:
a) Proportional to the weight of the body b) Proportional to the depth of submergence
c) **Equal to the weight of the fluid displaced** d) Independent of the volume of the body
85. The amount of company's profits that the board of directors of the corporation decides to distribute to ordinary shareholders.
a) Dividend b) Return c) Share of stock d) Par value
86. Which of the following is not a theorem on limits?
a) The limit of the algebraic sum of several functions is equal to the sum of their limits
b) The limit of the product of several function is equal to the product of their limits
c) The limit of the difference of several functions is equal to the difference of their limits
d) The limit of the quotient of two functions is equal to the quotient of their limits, provided the denominator is not zero
87. When a homogeneous, flexible cord is held at the two ends and allowed to sag freely on its own weight, it will produce a curve very similar to a parabola opening upwards. How do you call this curve?
a) Parabola b) **Catenary** c) Cycloid d) Epicycloids
88. Which of the following Common wealth Acts is known as the oldest Mechanical Engineering law?
a) Commonwealth Act 394 b) Commonwealth Act 594 c) **Commonwealth Act 294** d) Commonwealth Act 8394
89. What is a borrower of a particular loan almost always required to do during repayment?
a) Pay exactly the same amount of interest each payment. b) **Repay the loan over an agreed-upon amount of time**
c) Pay exactly the same amount of principal each payment d) The choices a and c above

90. To be a member of the Board of Mechanical Engineering, he/she must be at least how many years old?
a) 25 years b) 30years c) **35 years** d) 40yeras
91. How do you call the line passing through the focus and perpendicular to the directrix of a parabola?
a. Latus rectum **b. Axis of the parabola** c. Transverse axis d. Major axis
92. What is the ratio of the distance between the foci to the distance between the vertices in either hyperbola or ellipse?
a. Eccentricity b. Latus rectum c. Variance d. Deviation
93. It is a statement that one mathematical expression is greater than or less than another. How do you call this?
a) Conditional expression **b) Inequality** c) Interval d) Domain
94. It is a method of depreciation where a fixed sum of money is regularly deposited at compound interest in a real or imaginary fund in order to accumulate an amount equal to the total depreciation of an asset at the end of the assets estimated life. How do you call this depreciation?
a) Straight line method b) Declining balance method c) SYD method **d) Sinking fund method**
95. What is an artificial expense that spreads the purchase price of an asset or other property over a number of years?
a) Depreciation b) Amnesty c) Sinking fund d) Bond
96. How do you classify this interest rate, which specifies the actual rate of interest on the principal for one year?
a) Nominal rate b) Rate of return c) Exact interest rate **d) Effective rate**
97. What type of curve is generated by a point, which moves in uniform circular motion about an axis, while traveling with a constant speed, V, parallel to the axis?
a) A cycloid b) An epicycloids c) A hypocycloid **d) A helix**
98. What do you call the possible outcome of an experiment?
a) A sample space b) A random point **c) An event** d) A finite set
99. How do you call a sequence of numbers where the succeeding term is greater than the preceding term?
a) Dissonant series b) Isometric series c) Convergent series **d) Divergent series**
100. A branch of mathematics which uses the properties of numbers by using symbols or letters to represent numbers in arithmetic operations which usually variables and unknown quantities which usually involves the use and rearranging or equations.
a. Trigonometry **b. Algebra** c. geometry d. calculus

THE FOLLOWING ITEMS ARE FOR YOUR READING AND FAMILIARIZATION

101. This is a series of sequential method for carrying out a desire procedure to solve problem.
a. Algorithm b. hypsogram c. logarithm d. angstrom
102. This is use for expressing wavelengths of light or ultraviolet radiation with a unit or length equal to 10^{-10} metre.
a. Mersenne number b. Midac c. Light year **d. Angstrom**
103. It refers to a straight line, which a curve approaches closely, but never meets or touches the curve.
a. Asymptote b. Directrix c. Latus rectum d. Line segment
104. It is a collection of numbers or letters used to represent a number arranged properly in rows and columns.
a. Determinant b. Matrix **c. Array** d. equation
105. It is a high – level programming language for the computer used to express mathematical and scientific problems in a manner that resembles. English rather than computer notations.
a. Algol b. Cobol c. Pascal d. Aldus
106. In any triangle, the length of a line which is equal to the square root of the sides adjacent to the point where this line started minus the product of the segments of the third side is known as:
a. Angle bisector b. Median c. Perpendicular bisector d. Trisector
107. The whole is greater than any one of its parts. This statement is known as:
a. Postulate **b. Axiom** c. Hypothesis d. Theorem
108. A perpendicular segment from a vertex of the triangle to the line containing the opposite side is known as:
a. Median **b. altitude** c. Angular bisector d. Perpendicular bisector
109. An S.I. unit of area equal to 100 sq. m.
a. Arc b. Acre c. Hectares d. Are
110. The angle that the line of sight to the object, makes with the horizontal, which is above the eye of the observer, is called as:
a. Angle of elevation b. Angle of depression c. Acute angle d. Obtuse angle
111. In complex algebra, we use a diagram to represent a complex plane commonly called as:
a. Venn diagram b. Histogram **c. Argand diagram** d. Funicular diagram
112. The area bounded by two concentric circles is called as:
a. Annulus b. Ring c. Disk d. Sector
113. A series of numbers in which each number or term is derived from the preceding number by adding a constant value to it is known as:
a. Geometric sequence **b. Arithmetic sequence** c. Analytical sequence d. Differential sequence
114. 10 to the negative power of 18 is the value of the prefix:

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a. Atto	b. Femto	c. Micro	d. Pico
115.A series of equal payments occurring at equal periods of time.			
a. Annuity	b. Sinking fund	c. Cash flow	d. annual cost
116.The ratio of annual sales to the average of assets used in producing these sales.			
a. Inventory turnover	b. Asset turnover	c. Operating – expense ratio	d. Quick ratio
117.Quick ratio is defined as the ratio of quick assets to the current liabilities, sometimes this is called:			
a. Acid test ratio	b. Debt ratio	c. Equity ratio	d. Current ratio
118. Form of business/company ownership:			
a. Partnership	b. Corporation	c. Single proprietorship	d. All of the above
119.It is defined to be any method of repaying a debt, the principal and interest included usually by a series of equal payments at periodic intervals of time.			
a. Amortization	b. Annuity	c. Deferred annuity	d. Preferred annuity
120.Grand total of the assets operational capability of a corporation.			
a. Authorized capital	b. investment	c. Earning value	d. Money market
121.The process of recording all the transactions of the company, which affect any investment of capital so that at any time the results of investment, may be known as:			
a. Bookkeeping	b. Balancing accounts	c. Profit & loss statement	d. Accounting
122.A system of units based on time, length and mass is called:			
a. Absolute system	b. Gravitational system	c. cgs system	d. mks system
123.A US unit capacity used to measure solids equal to 7056 cubic inches (0.1156 m ³)			
a. Furlong	b. Barrel	c. Bushel	d. Bar
124.It refers to a statistical distribution having two distinct peaks of frequency distribution.			
a. Bimodal	b. Binomial	c. Biaxial	d. Bilingual
125. The straight line or plane that divides a line, a plane, or an angle into two equal parts.			
a. Divisor	b. Trisector	c. Eliminator	d. Bisector
126.A law, which states that the curvature of the central fiber is proportional to the bending moment in homogenous bar.			
a. Euclid	b. Thailus	c. Bernoulli – Euler	d. Torrecilli
127.A mathematical method that combines two numbers, quantities, etc., to give a third quantity. An example is the multiplication of two numbers in arithmetic.			
a. Binary Operation	b. Polynomial	c. Trinomial	d. sequence
128.An algebraic expression having two variables in it. For example, 3x+y is called			
a. Binary	b. Binomial	c. Polynomial	d. Trinomial
129.An algebra of sets that has two binary operations called addition and multiplication which may be used to represent binary logic is called:			
a. Boolean Algebra	b. Elementary Algenra	c. Matric	d. Laplace
130.A logarithm having a base 10 is called:			
a. Natural logarithm	b. Briggsian logarithm	c. Complex ogarithm	d. Naperian logarithm
131.Any two points along a streamline in an ideal fluid in steady flow, the sum of the pressure, the potential energy per unit volume and the kinetic energy per unit volume has the same value. This concept is known as the:			
a. Pascal's Theorem	c. Fluid Theory	c. Hydraulic Theorem	d. Bernoulli's Energy Theorem
132.If the temperature of a confined gas does not change, the product of the pressure and volume is constant. This is known as:			
a. Boyles Law	b. Pascal's Law	c. Archimedes principle	d. Torrecillis's Principle
133.The quantity of heat required to raise the temperature of one pound mass of water through one degree Fahrenheit is know as:			
a. Calorie	b. specific heat	c. British thermal unit	d. Energy